Answer the questions in the spaces provided on the question sheets. If you run out of room for an answer, continue on the back of the page.

Name: _____

Student ID: _____

Quiz 1: 09/08 Content

MATH 1271

09/10/2026

This quiz is worth 5 points and consists of 1 questions. You have 5 minutes to complete the quiz.

Question	Points
1	5
Total:	5

1. (5 points) Determine all of the roots of the function $f(x) = 6x^4 - 5x^3 - 4x^2$.

Solution: We find roots via factorization:

1. First we note all terms have a common factor of x^2 - so we factor that from the polynomial to get
$$x^2(6x^2 - 5x - 4) \tag{1}$$

2. Since x^2 is already factored into linear terms, we need only focus on factoring $6x^2 - 5x - 4$. This can be done either by applying the quadratic formula:
$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{5 \pm \sqrt{(-5)^2 - 4 \cdot 6 \cdot (-4)}}{2 \cdot 6} \tag{2}$$
$$= \frac{5 \pm \sqrt{25 + 96}}{12} \tag{3}$$
$$= \frac{5 \pm \sqrt{121}}{12} \tag{4}$$
$$= \frac{5 \pm 11}{12} \tag{5}$$
so our roots are $x = \frac{5+11}{12} = \frac{16}{12} = \frac{4}{3}$ and $x = \frac{5-11}{12} = \frac{-6}{12} = -\frac{1}{2}$. This implies that $(6x^2 - 5x - 4) = 6 \cdot (x - \frac{4}{3}) \cdot (x + \frac{1}{2}) = (3x - 4)(2x + 1)$.

   Alternatively, we note that factoring as $(ax + b)(cx + d)$ would force

   (a) $ac = 6$

   (b) $bd = -4$

   (c) $ad + bc = -5$.

   so we have $(a, c) = (6, 1), (3, 2), (-1, -6), (-2, -3)$ and $(b, d) = (4, -1), (2, -2), (1, -4)$. The only pair where $ad + bc = -5$ here would be $(a, c) = (3, 2)$ and $(b, d) = (-4, 1)$ giving us the above factorization.

3. We then have that
$$f(x) = 6x^2(x - \frac{4}{3})(x + \frac{1}{2}) \tag{6}$$
   from which we can read out the roots.

The roots of $f(x)$ are $x = 0, x = \frac{4}{3}$, and $x = -\frac{1}{2}$.